

Balancing Professionalism and Popularity: A Data-Driven Redesign of Elimination Rules

Summary

Dancing with the Stars faces controversy over its elimination rules, so we propose a “predict–compare–validate–optimize” framework to assess and refine the mechanism.

For Task 1, We used an improved **AGA-optimized BP neural network** to predict fan votes. Establishing the BP neural network architecture and a comprehensive loss function, and incorporating the genetic algorithm, our trained model achieved over 70% consistency in predicting eliminated contestants compared to actual eliminations, fulfilling the prediction requirement. The model’s stability was further validated through confidence and consistency tests, with both rules meeting over 75% compliance rates.

For Task 2, we first validated the hypothesis of significant divergence between the two methods **via chi-square tests**. Then, through normalization and defining critical weeks, we calculated **Spearman’s rank correlation coefficient** to compare method biases and identify controversial contestants. Graphical comparisons led to the conclusion: **the ranking method remained balanced across all periods without favoring any group, while the percentage method generally favored judges but shifted toward audience preferences during critical weeks**. Simulations were also conducted to compare the two methods and validate the effectiveness of the bottom-2 rule.

For Task 3, we first applied **PCA** to reduce three dancer-feature components into a composite score. We then built three **MLR models** and conducted F-tests, showing that dancer ability has a stable and significant positive effect on both competition outcomes and judge scores, while personal background has limited impact. Age and occupation mainly influence fan votes, highlighting inconsistent mechanisms between judges and audiences.

For Task 4, we use the **WOA-optimized nonlinear programming** obtained the **dynamic weighting rule**: fan influence leads early, while judge weight rises later to ensure balance. Table 9 shows this mechanism improves both entertainment and professionalism, yielding fairer eliminations across the season.

Sensitivity tests confirm that both the AGA-BP network and the whale regression scoring rule are robust, maintaining stage-dependent fan–judge weighting and stable overall performance

A final memorandum is provided to DWTS producers, highlighting the impact of vote-weighting schemes and suggesting refinements for upcoming seasons.

Keywords: AGA-BP network, Spearman’s rank, PCA, MLR, Dynamic Weighting

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1 Introduction

1.1 Problem Background

"Dancing with the Stars" is a celebrity dance competition where eliminations depend on both judges' scores and viewer votes, making fair score integration essential. Over 34 seasons, the show has used ranking-based and percentage-based consolidation rules, with Season 28 adding a Bottom-2 judge elimination. However, hidden vote data has raised ongoing fairness concerns, as popular contestants with low judge scores have sometimes advanced or won (e.g., Bobby Bones in Season 27).

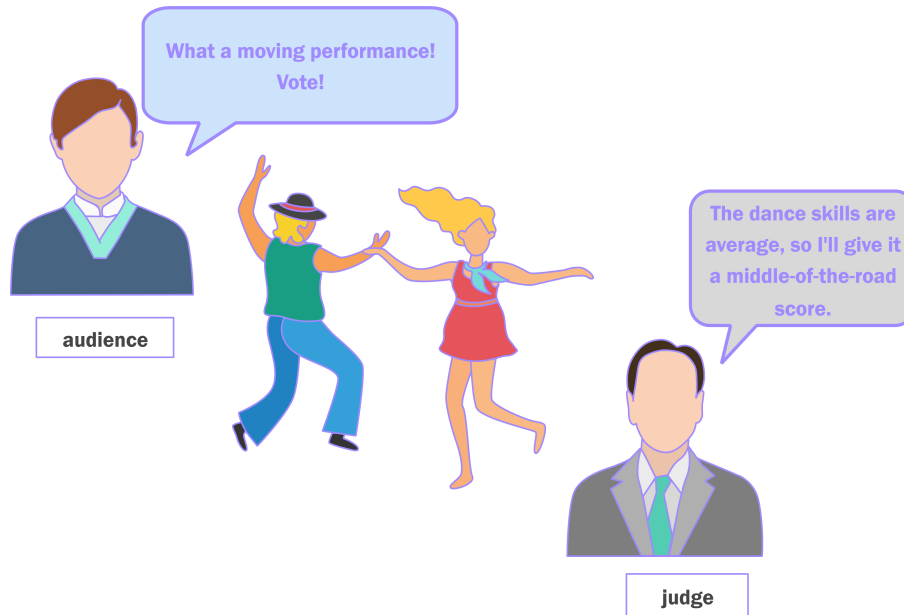


Figure 1: Audience and Jury Reviews

These evolving rules and recurring controversies highlight the difficulty of designing a voting system that balances fairness and audience engagement.

1.2 Restatement of the Problem

Our current objective is to analyze the voting and elimination system of Dancing with the Stars using the provided multi-season judge scores, contestant information, and final ranking dataset. The specific issues are as follows:

- **Task 1: Estimate the number of fan votes:**

By establishing a mathematical model to estimate the total number of fan votes each contestant receives weekly during the competition period.

- Can we predict the correct eliminations each week?
- How to estimate voting uncertainty

- **Task 2: Compare two official elimination rules:**

For both the cumulative ranking elimination rule and the percentage-weighted elimination rule

- Which one is more audience-friendly?
- Whether the results for the disputed athletes will be changed

- **Task 3: Analyze the Impact of Characteristics Between Contestants and Professional Dancers**

Establish a model for research purposes:

- Factors such as age, occupation, and dance partner affecting the competition
- Are the judges' scores consistent with the audience vote?

- **Task 4: Establish a new system that is more equitable.**

1.3 Our Work

On the basis of the above analysis, we have carried out our work. Our working framework is shown in Figure 2.

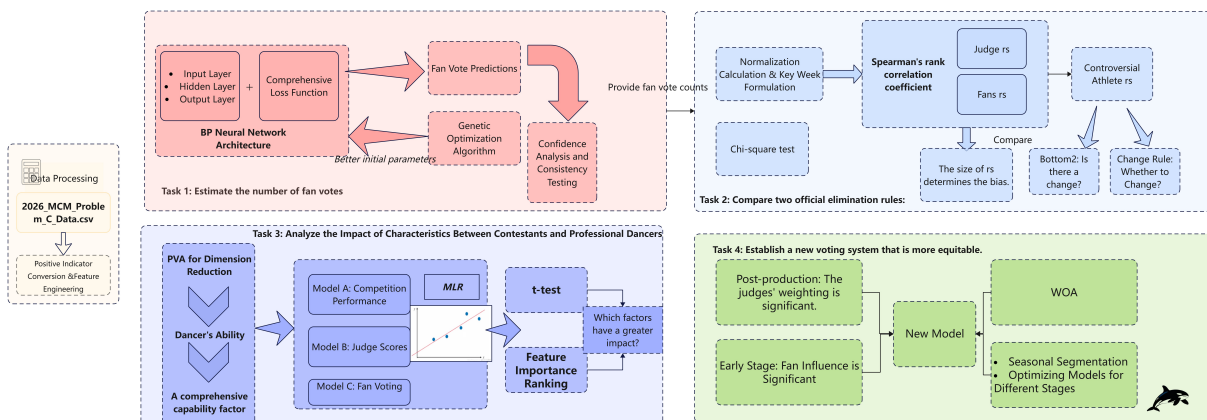


Figure 2: Our Work

The detailed modeling process and model details are presented in the following article.

2 Assumptions and Justification

- **Assumption:** In case of a tie, judges decide first, then the audience.

Because it is fundamentally a competitive event, professionalism comes first.

- **Assumption:** each week receives 10 million total votes per season.

Since fan vote totals are not disclosed, we assume an overall count.

- **Assumption:** dancers' skill levels remain constant.

simplify the model by excluding performance fluctuations due to partner skill changes.

3 Notations

Symbol	Description
i	Contestant index
t	Episode/time index
$\mathbf{x}_i$	Feature vector of contestant i
$y_i, \hat{y}_i$	True and predicted fan influence score
f_θ	Neural network prediction model
L	Total loss function
W_{judge}, W_{fan}	Dynamic judge/fan weights ($W_{judge} + W_{fan} = 1$)
S	Final combined score
rs_{fan}, rs_{judge}	Consistency correlations with fan/judge ranking

Table 1: Key notations used throughout the paper.

4 Data Pre-processing

2. Data Preprocessing and Feature Engineering

To ensure the DWTS dataset is suitable for statistical modeling, we implemented a streamlined preprocessing and feature engineering pipeline.

2.1 Performance Transformation

Contestant performance was transformed from ordinal rankings into a positively oriented metric:

$$\text{performance} = \text{max_rank} - \text{actual_rank} + 1$$

This transformation ensures that higher values represent better outcomes, aligning with modeling conventions.

2.2 Key Celebrity Attributes

We extracted key features that may influence audience voting behavior:

- **Region:** To avoid high-dimensional sparse vectors, U.S. states were grouped into six macro-regions—West, Midwest, South, Northeast, Overseas, and Unknown—and encoded using one-hot vectors.
- **Industry Background:** Categorized into Sports, Music, Film/TV, Reality Show, and Other, reflecting differences in fan base size and media exposure.
- **Age Group:** Divided into three brackets—20–30, 31–45, and 46+, capturing generational voting tendencies.

Table 2: Industry Category Distribution

Category	Count
Sports	112
Music	86
Film/TV	97
Reality Show	54
Other	72

Table 3: Home State Distribution

State	Count
California	72
New York	47
Texas	25
Florida	20
Georgia	15

Table 4: Age Group Distribution

Age Group	Count
20–30	127
31–45	175
46+	119

2.3 Professional Dancer Attributes

In addition to contestant-level features, we incorporated partner-related variables to reflect professional support and experience:

- **Participation Count:** Total number of seasons the dancer has participated in, representing cumulative experience.
- **Historical Appearances:** Number of prior appearances, used as a proxy for competition exposure.
- **Historical Scores:** Average scores across previous seasons (excluding competitive bias), reflecting professional credibility.

These features capture both technical expertise and the stabilizing influence of experienced partners on contestant performance.

2.4 Feature Scaling

Due to significant variation in the magnitude of input variables (e.g., scores, counts, rankings), all continuous predictors were standardized using z-score normalization:

$$x' = \frac{x - \mu}{\sigma}$$

This transformation ensures consistent feature ranges and improves the stability of neural network optimization.

2.5 Final Feature Integration

After preprocessing, the final dataset integrates:

- Demographic and performance indicators of contestants
- Encoded occupational and regional attributes
- Partner experience and historical scoring features
- Standardized continuous variables

This unified feature set provides a robust foundation for predictive modeling and performance evaluation.

5 Task 1: Estimate the number of fan votes

In "Dancing With The Stars", eliminations depend on judges' scores and fan votes, but vote totals are hidden. Thus, the key challenge is inferring fan support without vote labels.

BP neural networks are widely used in machine learning, featuring forward information propagation and backward error backpropagation.

Given that BP networks perform well on nonlinear data, we employ them to predict fan votes based on multi-season contestant features.

Traditional BP networks may get trapped in local minima, so we use a genetic algorithm to optimize initial weights and biases for better training accuracy.

Overall Model Framework

The overall architecture of this model is analyzed as follows:

- Data Input Module: Constructs a multidimensional feature vector containing player attributes, scoring information, and match progress.
- BP Neural Network Module: Performs forward propagation and error backpropagation.
- Adaptive Genetic Algorithm Module: Optimizes initial weights and thresholds to enhance model convergence and prediction accuracy.
- Output Module: Generates fan influence values or ranking probabilities for subsequent elimination decisions and mechanism analysis.

5.1 BP Neural Network Architecture

To capture the nonlinear relationship between fan voting behavior and contestant performance, we constructed a BP neural network as the core predictor.

1. Input Layer

The input dimension d equals the size of the feature vector. Meaning

$$d = \dim(X_{i,t})$$

In our implementation, each contestant is represented by the following feature vector:

- Judge Score Statistics: Weekly average score, standard deviation
- Competition Status: Weeks completed, current week's participants (rank method neural input proprietary), average score ranking/percentage
- Contestant Information: Age, industry, home state (non-US states uniformly labeled as foreign)
- Historical Scores: Average scores from previous weeks

2. Hidden Layer

We adopted a two-hidden-layer architecture to balance the model's expressive power with computational efficiency.

- First hidden layer: The first hidden layer has 14 ReLU neurons, capturing primary feature interactions and initial abstractions
- Second hidden layer: The second hidden layer has 7 neurons with ReLU activation, enabling nonlinear feature extraction and reducing vanishing gradients: $\text{ReLU}(x) = \max(0, x)$.

3. Output Layer

The output consists of individual neurons, each representing a contestant's raw fan influence score. This output is subsequently normalized weekly to generate a fan vote metric prediction with a total sum of 1.0.

Mathematical representation as follows: Forward propagation equation:

$$\begin{aligned}
 \mathbf{z}^{(1)} &= \mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)} \\
 \mathbf{a}^{(1)} &= \text{ReLU}(\mathbf{z}^{(1)}) \\
 \mathbf{z}^{(2)} &= \mathbf{W}^{(2)}\mathbf{a}^{(1)} + \mathbf{b}^{(2)} \\
 \mathbf{a}^{(2)} &= \text{ReLU}(\mathbf{z}^{(2)}) \\
 \hat{y} &= \sigma(\mathbf{W}^{(3)}\mathbf{a}^{(2)} + b^{(3)})
 \end{aligned} \tag{1}$$

Where $\mathbf{W}^{(i)}$ and $\mathbf{b}^{(i)}$ denote the weight matrix and bias vector of layer i , respectively; $\mathbf{x}$ is the input feature vector; and y is the predicted fan influence score.

Number of parameters: The network configuration utilizes 80 features, yielding approximately 1,600 parameters.

Ultimately, our neural network model was constructed as follows:

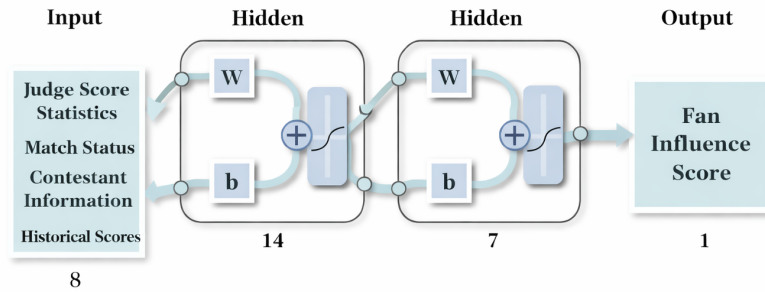


Figure 3: Neural Network Architecture

5.2 Loss Function Design

Our loss function is specifically designed for the unique format of the DWTS competition, particularly addressing the constraint that eliminated contestants must rank lowest in the combined fan-judge scoring system. To this end, we employ a loss function incorporating two rules:

Mean Square Error Rule (MSE)

$$L_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2$$

The baseline MSE measures the prediction accuracy of pseudo labels generated from historical data processing derived from two elimination patterns: where N is the number of contestants in a given week, y_i is the predicted fan vote count for contestant i , and $\hat{y}_i$ is the label fan vote count calculated using exponential smoothing based on their judge rankings.

Elimination Ranking Penalty

To satisfy the constraint that the eliminated contestant must have the worst overall score, we introduce a margin-based ranking loss: Let the eliminated contestant be indexed as e and the safe contestant as j .

For the percent elimination system: In penalty predictions, the total score (judges + fans) of eliminated contestants should be higher than that of all remaining safe contestants.

$$L_{\text{Rank}}^P = \sum_e \sum_{j \neq e} \max(0, (J_e + \hat{s}_e) - (J_j + \hat{s}_j) + \alpha)$$

For the Rank system: A penalty occurs when the eliminated contestant's "Composite

Ranking'' (judges' ranking percentage + (1 - fan share)) is lower than any safe contestant's.

$$L_{\text{Rank}}^R = \sum_e \sum_{j \neq e} \max(0, (R_j + (1 - \hat{s}_j)) - (R_e + (1 - \hat{s}_e)) + \alpha)$$

Among these, J represents the normalized share of the judge's score, R denotes the judge's ranking percentage, $\alpha > 0$ λ is the marginal parameter, and λ is the weight coefficient of the penalty term.

Considering the differences in voting mechanisms across seasons, we apply differentiated penalty weights:

Ranking-based seasons (1, 2, 28-34): Weighted at rank=5.6 or higher to reflect the discrete nature of the ranking system, where fan rankings are directly combined with judge rankings.

Percentage-based seasons (3-27): Weighted at percent=3.3 to account for the aggregate of continuous voting percentages.

$$L = L_{\text{MSE}} + \lambda \cdot L_{\text{Rank}}$$

In this approach, the weighting factor λ is selected differently for each season. This design ensures that our model not only fits the data effectively but also strictly adheres to the hard constraints imposed by the actual elimination outcomes.

5.3 Adaptive Genetic Algorithm

Gradient methods struggle with our non-convex constraints, so we use an adaptive GA for global search, then BP for local refinement, combining exploration with efficient convergence.

5.3.1 Genetic Algorithm Design

- **Representation method:** Each individual in the population is a flattened vector 0 containing all neural network parameters (weights and biases).
- **Population size :** 80 individuals provide sufficient diversity for exploration.
- **Initialization:** Parameters are sampled from $N(0, 0.3)$ to ensure diverse starting points.
- Fitness function

$$f(\theta) = \frac{1}{L_{\text{total}}(\theta) + \varepsilon} \quad \text{where} \quad \varepsilon = 10^{-\delta}$$

- Algebra

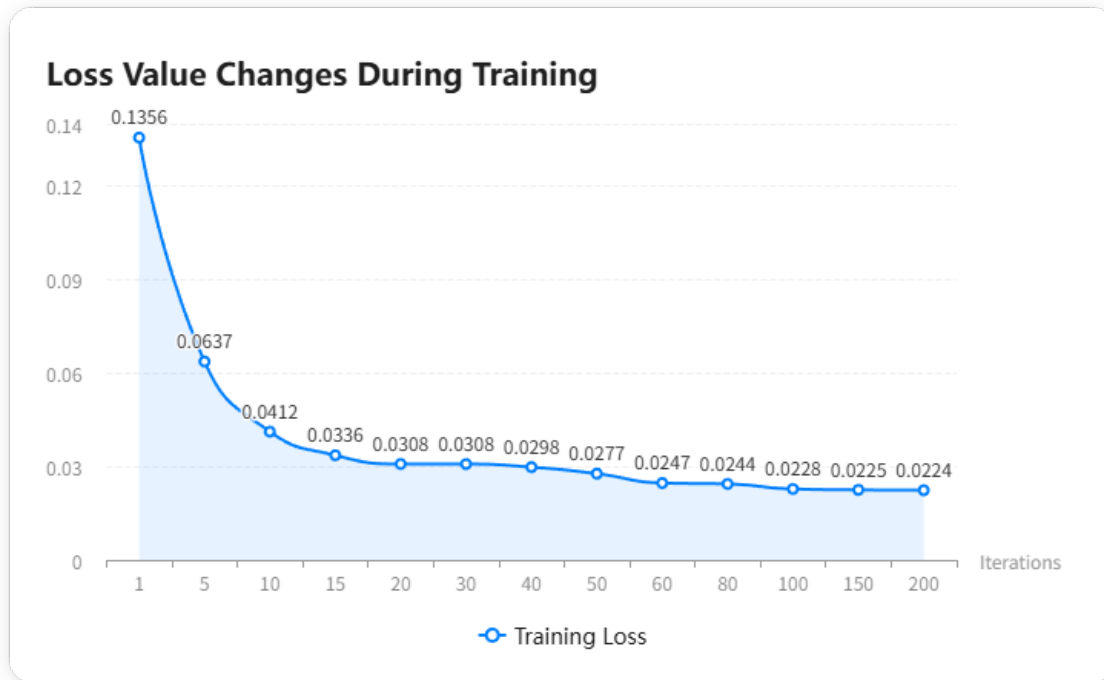


Figure 4: Loss Value Changes During Training

The training loss decreases significantly as iterations increase, stabilizing after a certain number of iterations and reaching a minimum of 0.0224. We ultimately selected 40 iterations to balance convergence quality and computational cost.

- **Selection Strategy** We employ k-tournament selection ($K=3$) to maintain selection pressure while preserving diversity. In each selection round, k individuals are randomly sampled, and the one with the highest fitness is chosen.

Parameter ranges are defined as $p_{cE} \in [0.6, 0.9]$ and $p_{mE} \in [0.005, 0.10]$. Individuals with high fitness are assigned lower mutation rates to preserve high-quality solutions, while those with low fitness receive higher mutation rates to help escape local optima.

5.4 Results Analysis

As shown in the figure, training and test accuracy fluctuate with loss, reaching 83.18% and 86%, with no clear upward trend. After GA optimization, the model achieved over 70% consistency with historical eliminations, reliably identifying the lowest-ranked contestants across weeks. This demonstrates that the model successfully captured the key factors influencing elimination, thereby fulfilling the prediction task outlined in the problem statement.

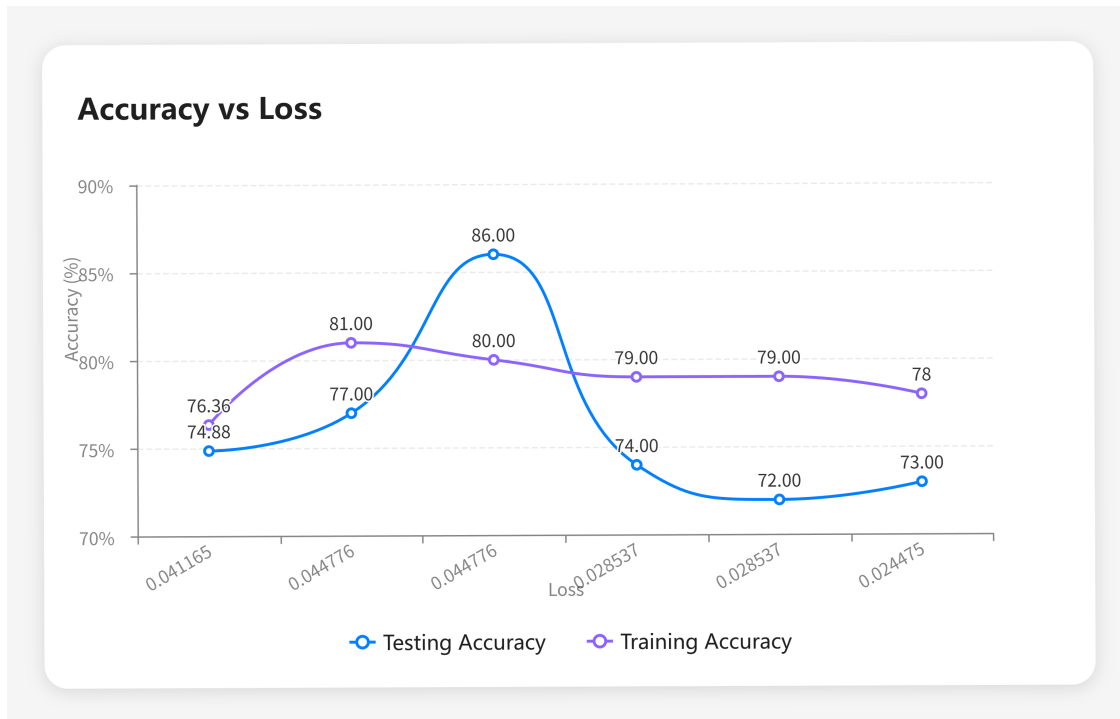


Figure 5: Accuracy vs Loss

5.5 Confidence Analysis and Consistency Testing of Prediction Results

We evaluate performance using RMSE and CV for error and volatility, compliance rates for elimination-rule consistency, and cohort reliability via industry consistency, weekly stability, and Mann–Whitney U tests.

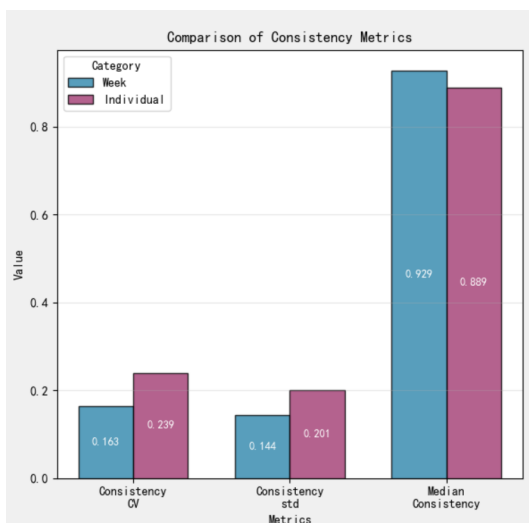


Figure 6: Comparison of Consistency Metrics

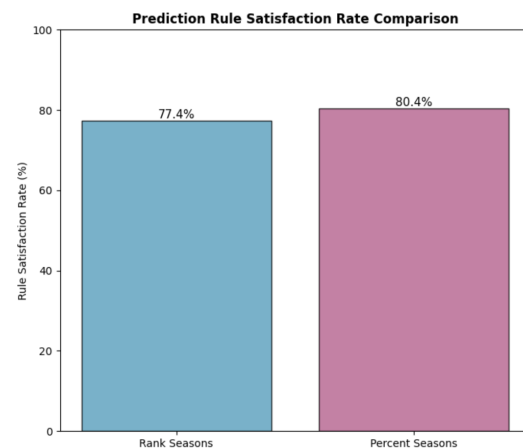


Figure 7: Prediction Rule Satisfaction Rate Comparison

As shown in the figure, weekly CV (0.1632) and Std (0.144) are lower than player-level

values, indicating stable performance over time but larger individual differences. Week and player consistency scores (0.9286, 0.8889) are both close to 1, confirming strong predictive reliability.

Figure 2 shows fulfillment rates of 80.4% in percentage seasons and 77.4% in ranking seasons, suggesting better adaptability under percentage rules. Overall, the GA-BP model achieves low error, high stability, and strong cross-season consistency, making it reliable for fan vote prediction and performance evaluation.

6 Comparing Two Official Elimination Rules

In Model 1, we inferred fan vote counts. Using these along with judge scores and elimination outcomes, Task 2 evaluates ranking vs. percentage systems via Spearman correlations, identifies controversial weeks and contestants with judge–fan divergence, and assesses how the Bottom-2 rule affects their advancement.

6.1 Chi-Square Test for Differences Between Two Elimination Methods

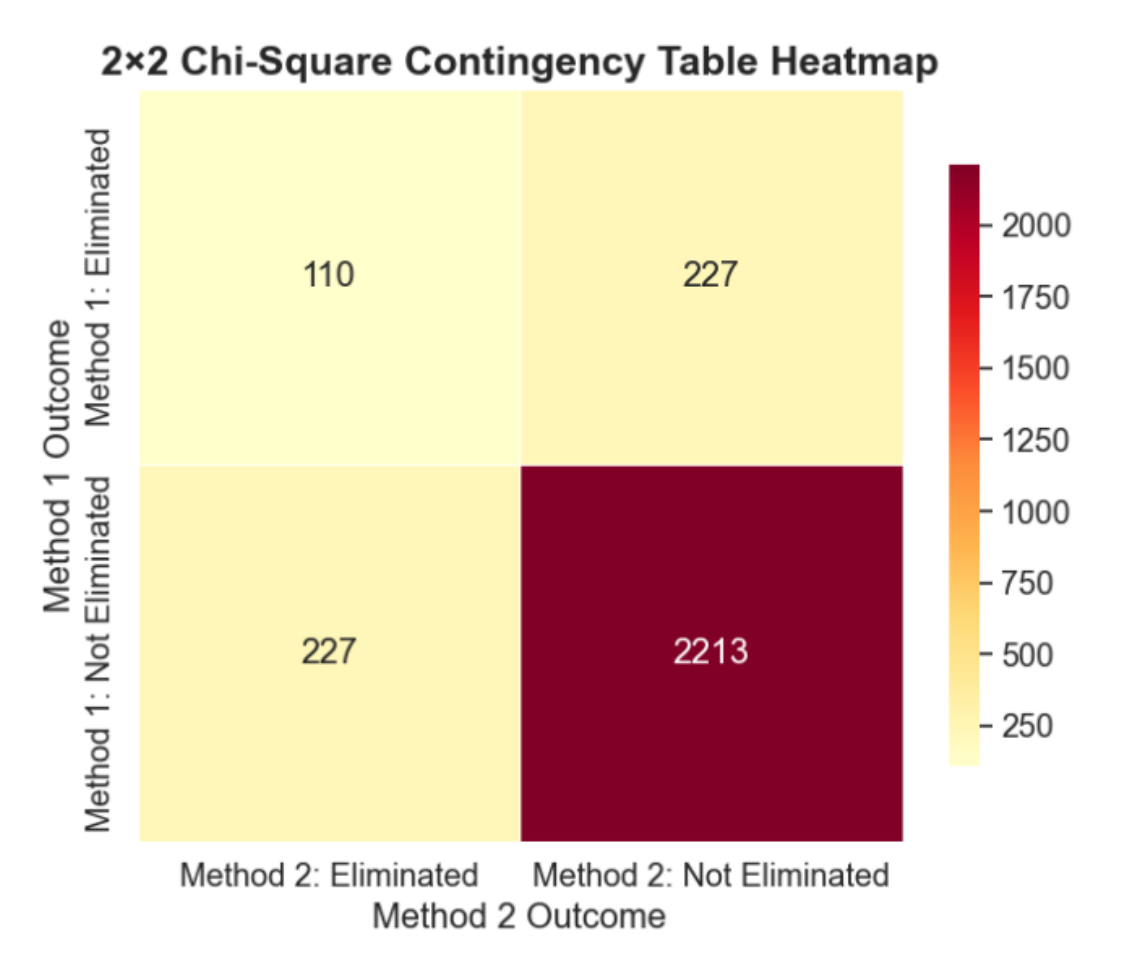


Figure 8: Chi-Square Contingency Table Heatmap

There is a highly significant difference between the two methods in terms of elimination targets, with a substantial divergence in outcomes (the overlap in eliminated individuals is approximately 19.5%).

6.2 Quantification of Rule Bias: Spearman's Rank Correlation

Create a data table

We built a structured dataset with key rule-related variables (judge totals, inferred fan votes, ranks, percentages, and elimination labels). This enables systematic comparison of outcomes under different rules and quantifies fan bias using Spearman's rank correlation.

Table 5: Derived Variable Definitions

Derived Variable	Calculation Logic
judge_total	Sum of valid judge scores for the player this week (ignore N/A)
judge_rank	Rank by judge total score (descending); ties share rank
fan_rank	Rank by audience votes (descending); ties share rank
rank_combined_rank	Sum of judge rank + fan rank; lower is better; judge rank breaks ties
rank_eliminate	Bottom k by combined rank are eliminated (based on result field)
judge_pct	Judge score percentage of total judge scores (see Appendix 2)
fan_pct	Vote percentage of total audience votes (see Appendix 2)
pct_combined_score	Sum of judge_pct + fan_pct (see Appendix 2)
pct_eliminate	Bottom k by combined score are eliminated

6.2.1 Definition of Bias

To evaluate the alignment between final rankings and different scoring sources, we compute the Spearman rank correlation coefficients:

$$rs_{\text{fan}} : \text{Final vs. fan rankings}, \quad rs_{\text{judge}} : \text{Final vs. judge rankings}.$$

A rule with a higher rs_{fan} indicates stronger alignment with fan preferences, whereas a higher rs_{judge} suggests judge-driven outcomes.

To account for potential fan influence in critical stages, we apply the *key week hypothesis*, in which the final 2–3 weeks are treated as key weeks. Spearman correlations are computed separately for these weeks to capture possible fan dominance caused by extreme voting behaviors.

Here we introduce two key data processing concepts.

- Normalization Module

Since our focus is not on mid-tier players, we normalize the data to emphasize high and low rankings using a nonlinear normalization function:

$$\text{normalized} = \frac{\tanh\left(2 \cdot \left(2 \cdot \frac{\text{rank}-1}{n-1} - 1\right)\right)}{\tanh(2)}$$

This approach downweights mid-range values. Here, *rank* is the original 1-based ranking, and *n* is the total number of competitors that week. A hyperbolic tangent function is used to magnify top-bottom differences (e.g., 1st vs. 3rd among three competitors outweighs 8th vs. 10th among ten).

- Time Weighting Module

We weight matches by week to reflect the greater importance of later competitions. The weight increases linearly from 0.5 in early weeks to 1.5 in later key weeks, amplifying the influence of decisive stages.

Table 6: Spearman Rank Correlation (r_s) Comparison Across Methods and Time Periods

Comparison Type	Method 1 (Rank)	Method 2 (Percent)
Final vs Fan (All Weeks)	0.224	-0.160
Final vs Fan (Key Weeks)	0.188	0.279
Final vs Judge (All Weeks)	0.279	0.422
Final vs Judge (Key Weeks)	0.294	-0.100

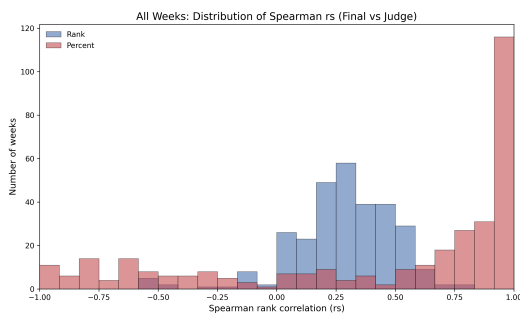


Figure 9: RS Distribution Histogram (Judge vs. Final) Full Week

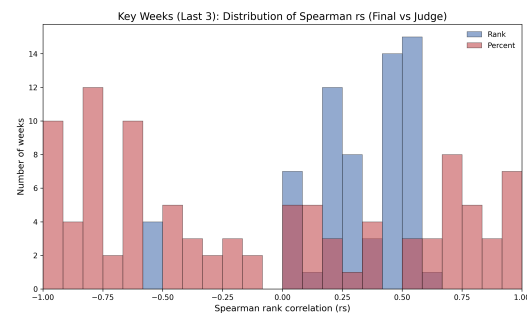


Figure 10: RS Distribution Histogram (Fans vs. Final) Crucial Week

From the RS comparison, we conclude

- Full-week rank method: Judge r_s (0.279) is close to fan r_s (0.224), indicating a relatively balanced method.
- Full-week percent method: Judge r_s (0.422) exceeds fan r_s (0.16), indicating a judge-favoring method with partial negative correlation to fans.

Refer to Figures 11 and 12.

- Critical week, rank method: $r_s = 0.294$ vs. fan $r_s = 0.188$, remaining relatively balanced.
- Critical week, percent method: $r_s = 0.100$ vs. fan $r_s = 0.279$, indicating the percent rule favors fans during critical weeks.

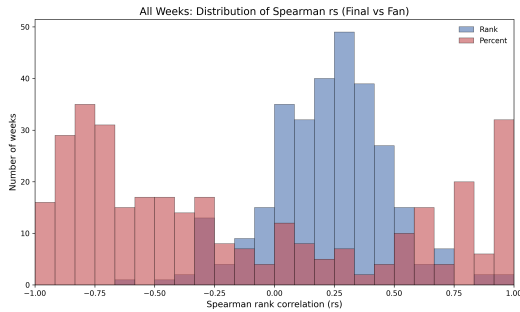


Figure 11: RS Distribution Histogram (Fans vs. Final) Full Week

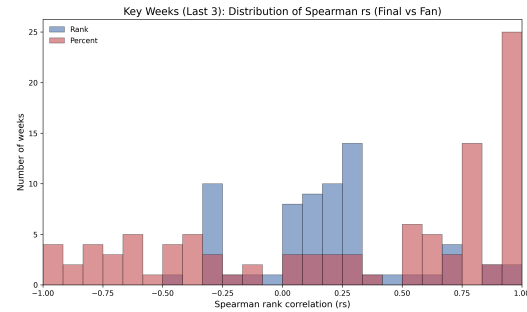


Figure 12: RS Distribution Histogram (Fans vs. Final) Crucial Week

Overall, the rank method is relatively balanced across both full and key weeks. In contrast, the percent method favors judges in full weeks but shifts toward fans in key weeks, where extreme voting can dominate.

6.2.2 Controversial Athlete

A contestant is defined as 'controversial' if both conditions hold: 1. Valid appearances 8 (eliminations excluded) 2. Personal $rs < 0.73$ (judges consistently rank lower than fans)

Using the normalized data, we computed weighted rs values and identified over 100 controversial contestants. Representative examples are shown in the chart.

Table 7: Spearman Coefficient, Seasons, and Representative Normalized Rankings

Contestant	rs (4 d.p.)	Season(s)	Judge (repr.)	Fan (repr.)
Billy Ray Cyrus	-0.9937	4	1.0000 (total 8)	0.0000 (total 8)
Bobby Bones	-0.9496	27	0.1977 (total 9)	0.6668 (total 9)
Bristol Palin	-0.8375	11, 15	0.5933 (total 14)	0.2422 (total 14)

Billy's rs of 0.9937 (near 1) makes him a prototypical controversial contestant. Judges rank him consistently low, while fans rank him high, marking him as a classic fan favorite.

On 100 disputed players, switching 'percent $\rightarrow$ rank' improved far more rankings than it worsened (48 vs. 6), while 'rank $\rightarrow$ percent' showed similar gains and losses (11 vs. 10). This indicates the rank method is more inclusive, whereas the percent method tends to suppress some players, supporting the fairness conclusion.

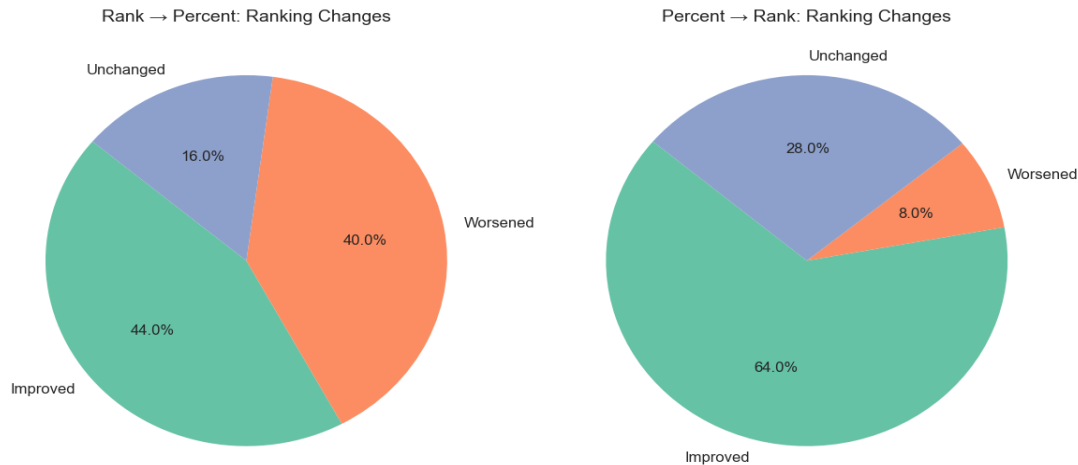
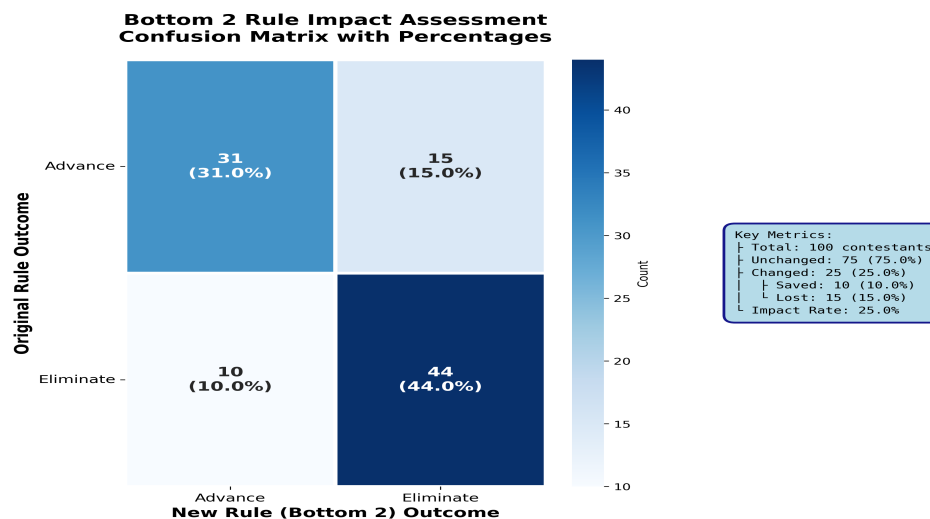


Figure 13: Exchange between percent and rank

We next introduce the 'Bottom 2' rule to examine its impact on disputed players. Under this rule, the two lowest combined rankings enter the Bottom 2, and the contestant with the lower judge ranking is eliminated (fan ranking breaks ties).

Figure 14: $bottom2_{confusion_{matrix}}$

The Bottom-2 rule changed the advancement outcomes for 25% of the disputed contestants ($\chi^2 = 18.73, p < 0.001$): 15 shifted from advancing to elimination and 10 from elimination to advancing, yielding a net change of -5 . This shows that the rule significantly, though moderately, affects fairness by strengthening judge influence while retaining fan input.

We recommend adopting the Bottom 2 rule selectively: inactive early, conditionally used

mid-season, and primarily applied in late stages to balance professionalism and entertainment.

7 Task 3: Analyze the Impact of Characteristics Between Contestants and Professional Dancers

In 'Dancing with the Stars', celebrity–dancer pairings strongly affect performance, judging, and audience support. A scientific strategy quantifies the effects of contestant traits and dancer abilities on multiple objectives. We therefore adopt the framework: 'PCA for dimension reduction → MLR + ANOVA for feature importance'.

7.1 PCA for Dimension Reduction

Principal Component Analysis of Dancer Ability

To evaluate dancer ability, we extracted three highly correlated indicators:

- Number of competition rounds
- Cumulative weeks participated
- Historical average score

The correlation coefficients among these variables ranged from 0.71 to 0.83, indicating strong linear relationships and shared underlying structure. Therefore, we applied Principal Component Analysis (PCA) to synthesize a unified ability metric.

PCA Results:

- The first principal component (`dancer_ability_pc`) explained **78.24%** of the total variance.
- Component loadings:
 - Competition rounds: 0.58
 - Cumulative weeks: 0.61
 - Historical average score: 0.54

Interpretation: The loadings are balanced across all three indicators, suggesting that the principal component effectively captures a composite of dancer seniority, technical proficiency, and endurance. A single component is sufficient to represent overall dancer ability for downstream modeling.

7.2 MLR+ANOVA

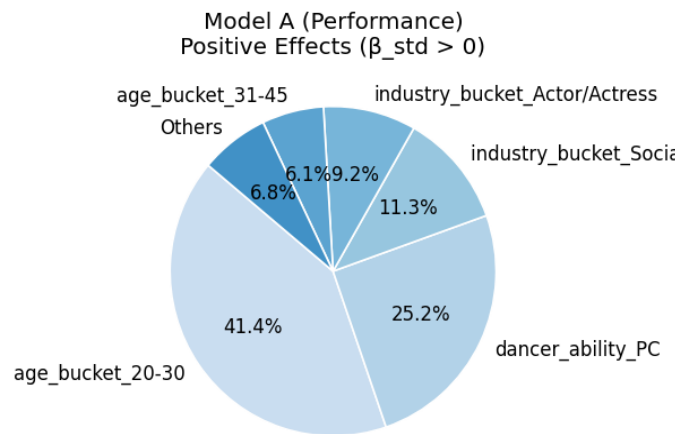
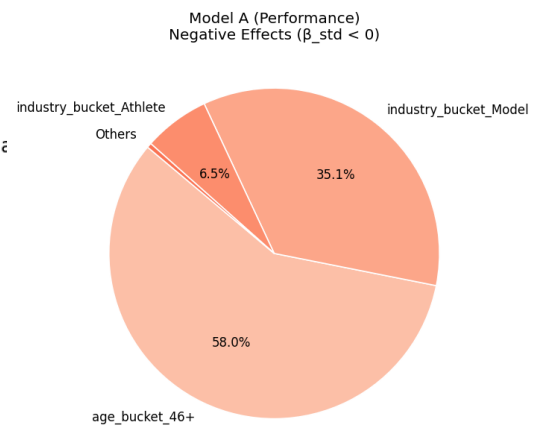
In 'Dancing with the Stars', celebrity–dancer pairings strongly affect performance, judging, and audience support. A scientific strategy quantifies the effects of contestant traits and dancer abilities on multiple objectives. We therefore adopt the framework: 'PCA for dimension reduction → MLR + ANOVA for feature importance'.

Table 8: Dependent Variables and Definitions for Models A–C

Model	Dependent Variable	Definition
Model A	performance	Standardized rank score; higher means better performance
Model B	judge_total_mean	Average judge score (range: 20–30)
Model C	fan_vote_mean	Predicted average fan vote share from Model 1

These three models examine factors affecting ‘final rankings’, ‘judges scores’, and ‘fan votes’, respectively.

After factor selection, we test whether these variables have real effects. Each regression applies F-tests to assess individual variable significance ($H_0: \text{coefficient} = 0, \alpha = 0.05$), retaining factors with $p < 0.05$. Significant variables are then ranked by the absolute standardized coefficients to compare relative importance across models.

Figure 15: Model A (Performance) Positive Effects ($\beta_{std} > 0$)Figure 16: Model A (Performance) Negative Effects ($\beta_{std} < 0$)

Model A reports $R^2 = 0.209$, $\text{Adj } R^2 = 0.190$, and an F-test of $p = 2.20 \times 10^{-16}$, confirming overall significance and indicating that athlete characteristics explain variation in outcomes. The results show that younger contestants (20–30) have a strong advantage, older contestants (46+) face a clear disadvantage, and those aged 31–45 are neutral. Modeling experience has a negative effect, whereas dancer ability exhibits a strong positive effect, supporting the validity of the PCA-derived indicators.

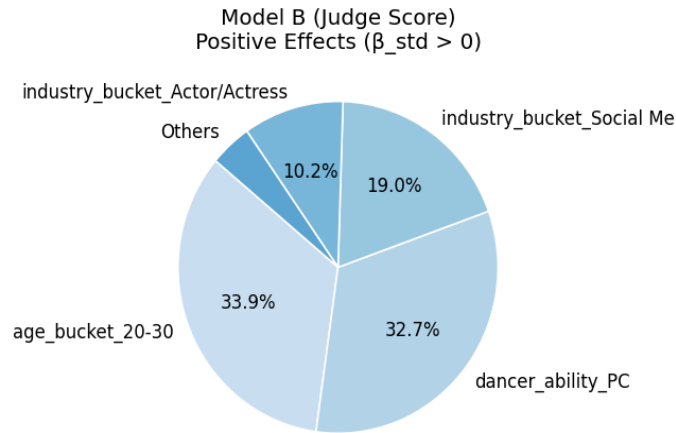


Figure 17: Model B (Judge Score) Positive Effects ($std > 0$)

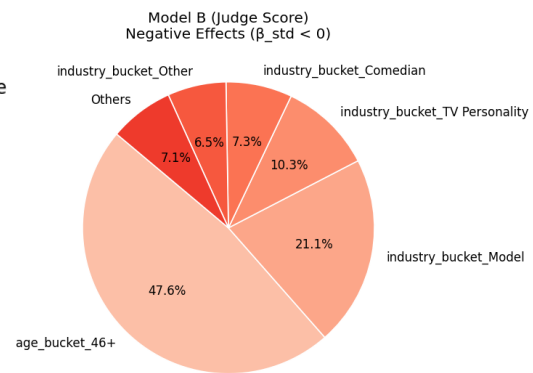


Figure 18: Model B (Judge Score) Negative Effects ($std < 0$)

Model B exhibits the highest explanatory power ($R^2 = 0.237$) with a highly significant F-test, indicating that judges' scoring behavior is strongly professional. The results are consistent with those of Model A: age and dancer ability positively influence judges' scores, whereas a modeling background has a negative effect. Judges also tend to favor performance-oriented professions, with social media professionals (19%) and actors (10.2%) contributing positively to the scores. Overall, the findings suggest that judges place substantial emphasis on technical skill, age, and stage presence.

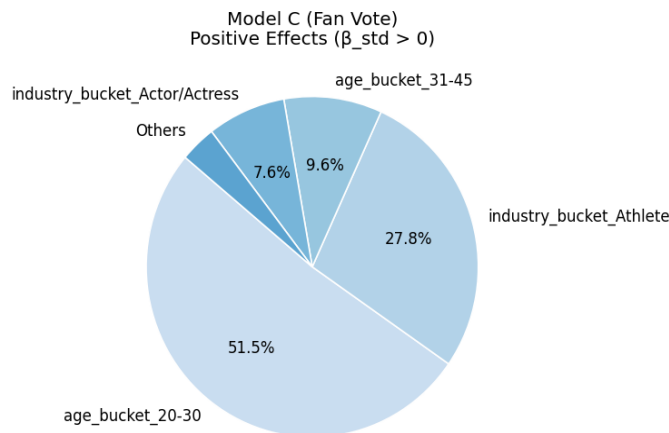


Figure 19: Model C (Fan Vote) Positive Effects ($std > 0$)

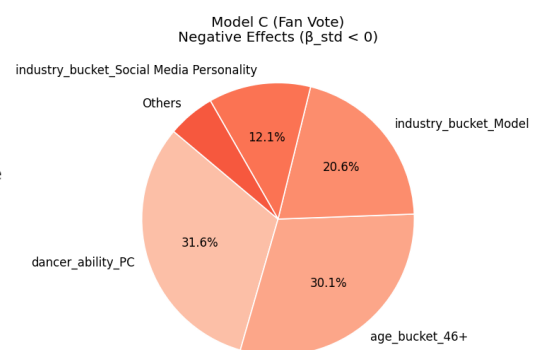


Figure 20: Model C (Fan Vote) Negative Effects ($std < 0$)

Model C demonstrates lower explanatory power ($R^2 = 0.122$, $p < 0.001$), indicating that fan voting is less structured and more strongly influenced by subjective factors such as popularity and social trends. Similar to the other models, younger contestants receive

higher support; however, unlike Models A and B, dancer ability contributes negatively (31.6%), suggesting that stronger technical skills may actually reduce fan votes.

Overall, dancer skill consistently enhances performance and judge scores, whereas personal background exerts limited influence. Age and profession primarily affect outcomes through fan voting. Judges place emphasis on professional competence, while fans integrate emotional preference and social appeal alongside skill.

8 Establish a new voting system that is more equitable.

Based on Question 2, the percent method is highly comprehensive, balancing judge professionalism with fan preferences and suiting mass-market entertainment. However, fixed weights may dampen early fan enthusiasm and allow late-stage fan dominance to undermine professionalism.

We therefore propose an improved percent-based system with '**dynamic weighting**': fan influence dominates early to boost engagement, judge influence increases later to ensure credibility, with a smooth mid-stage transition. Weights are adaptively optimized by an algorithm rather than manually set.

8.1 Three-Stage nonlinear programming Model

Let W be the total number of competition weeks. The season is divided into three stages: early ($F = \lceil W/3 \rceil$), middle ($F = \lceil (W - F)/3 \rceil$), and late ($W - F_{\text{early}} - F_{\text{middle}}$). For example, $W = 8$ yields (3, 3, 2); $W = 9$ yields (3, 3, 3).

The scoring formula is:

$$S = W_{\text{judge}} \cdot P_{\text{judge}} + W_{\text{fan}} \cdot P_{\text{fan}}, \quad W_{\text{judge}} + W_{\text{fan}} = 1, \quad W_{\text{judge}}, W_{\text{fan}} \geq 0$$

where P_{judge} and P_{fan} are percentage scores from judges and fans, respectively.

Early Stage Optimization

$$\max rs(\text{Rank}_{\text{fan}}, \text{Rank}_{\text{final}})$$

Subject to:

$$W_{\text{judge}} + W_{\text{fan}} = 1, \quad W_{\text{judge}}, W_{\text{fan}} \geq 0, \quad rs_{\text{judge}} \geq rs_{\text{judge}}^* = 0$$

Goal: prioritize fan influence to boost engagement, while maintaining minimal judge input for credibility.

Middle Stage (Fixed Weights) Set $W_{\text{judge}} = W_{\text{fan}} = 0.5$ to ensure stability and avoid abrupt shifts. This balances emerging contestant strength with fan base consolidation.

Late Stage Optimization

$$\max rs(\text{Rank}_{\text{judge}}, \text{Rank}_{\text{final}})$$

Subject to:

$$W_{\text{judge}} + W_{\text{fan}} = 1, \quad W_{\text{judge}}, W_{\text{fan}} \geq 0, \quad r s_{\text{fan}} \geq r s_{\text{fan}}^* = -0.2$$

Goal: emphasize professional judgment in final outcomes while preserving minimal fan influence.

Whale Optimization Algorithm

Optimizing the weights ($W_{\text{judge}}, W_{\text{fan}}$) with the constraint $W_{\text{judge}} + W_{\text{fan}} = 1$, $W_{\text{judge}}, W_{\text{fan}} > 0$ forms a nonlinear constrained problem where gradient-based methods fail. We therefore adopt an improved Whale Optimization Algorithm (WOA) for its strong global search ability and stable convergence. WOA (Mirjalili, 2016) is a swarm-based metaheuristic inspired by humpback whale hunting behavior and consists of three core phases:

1. Encircling prey: move toward the current best solution;
2. Bubble-net attack: spiral contraction balancing exploration and exploitation
3. Random search: random movement to escape local optima.



Figure 21: WOA

8.2 Data Analysis

Table 9: Average Metrics by Stage

Stage	W_{judge}	W_{fan}	$r s_{\text{fan}}$	$r s_{\text{judge}}$
Late Stage	0.6262	0.3738	-0.0875	0.4234
Early Stage	0.2682	0.7318	0.4387	0.0096

Stage-Based Voting Mechanism Validation

In the late stage, the optimized weights satisfy $W_{\text{judge}} > W_{\text{fan}}$, confirming stronger judge influence. In contrast, early-stage optimization consistently yields $W_{\text{fan}}^* > W_{\text{judge}}^*$, enhanc-

ing fan impact and audience engagement.

To validate late-stage objectives, judge consistency improves significantly:

$$rs_{\text{judge}} \approx 0.30 \text{ to } 0.40$$

while maintaining minimum fan satisfaction:

$$rs_{\text{fan}} \geq 0.1$$

Overall, the proposed mechanism achieves:

- Fan representation in early stages
- Professional fairness in late stages
- Safeguards against unilateral dominance
- Stable optimal weights via WOA optimization

This stage-wise balance enhances the fairness and credibility of the elimination process.

9 Sensitivity Analysis

Table 10: With a lower threshold

Stage	W_{judge}	W_{fan}	rs_{fan}	rs_{judge}	rs_{fan}^*	rs_{judge}^*
Late Stage	0.6799	0.3201	-0.2983	0.6108	-0.4	-0.2
Early Stage	0.2244	0.7756	0.6202	-0.1944	-0.4	-0.2

As shown in our whale regression model, reducing iterations and population size has little effect on the results. After reducing $rs^{\hat{*}}$, we obtained more extreme weights and rs values, which was undesirable. Following multiple rounds of debugging, we finalized the values for $rs^{\hat{*}}$ as -0.2 and 0.

Hyperparameter Sensitivity Analysis

We performed sensitivity analysis on neural network models, evaluating robustness via composite scores after iterative hyperparameter tuning.

We conducted a sensitivity analysis on key hyperparameters to evaluate their impact on model performance. The composite sensitivity scores are shown in the figure:

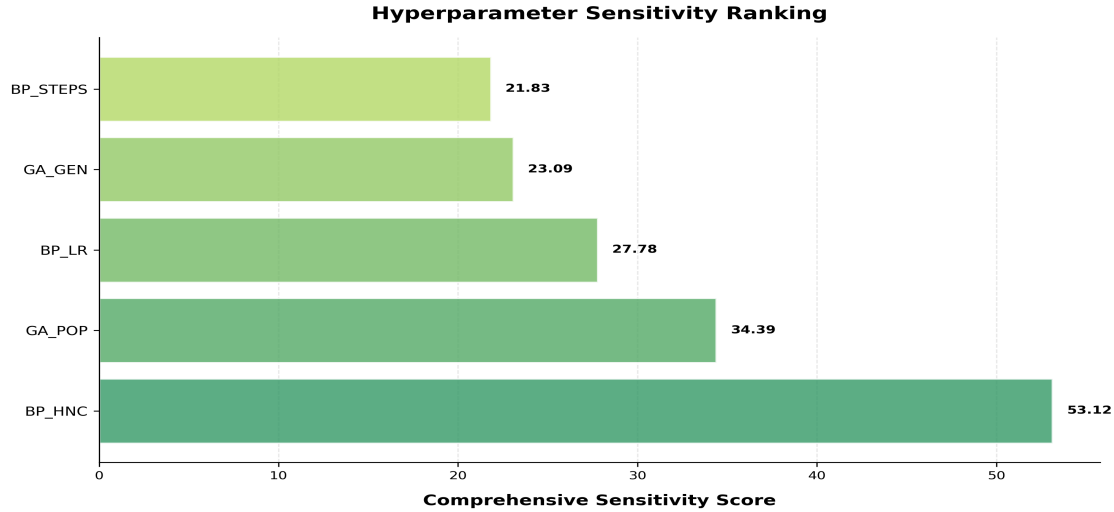


Figure 22: Hyperparameter Sensitivity Ranking

Overall, the sensitivity analysis provides clear interpretability and reliable guidance for hyperparameter optimization, contributing to improved model stability and generalization.

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CONCLUSION AND RECOMMENDATION

To: MCM office

From: MCM Team 2607943

Subject: MCM

Date: February 3, 2026

We are pleased to provide you with a summary that *Dancing with the Stars* faces ongoing controversy over elimination rules that struggle to balance entertainment and professionalism. This study analyzes their alignment with fan and judge preferences over time and proposes a 'dynamic weighting + percentage' elimination method.

Using preprocessed data, we applied a BP neural network optimized by an adaptive genetic algorithm to predict fan votes under both Rank and Percent methods. Both approaches achieved over 75% rule compliance. Simulated weekly rankings were evaluated using Spearman's correlation, leading to the following insights:

- The Rank method offers a relatively balanced outcome.
- The Percent method favors judges in early stages and fans in later stages, likely due to amplified fan extremes.
- The Bottom 2 rule effectively removes controversial contestants but overly favors judges, potentially reducing entertainment value.

Regression analysis reveals that age is the strongest predictor of performance—older contestants tend to perform worse. Judge scores correlate positively with professional dancer skill, while fans prefer athletes, reflecting a contrast between technical and subjective preferences.

We recommend adopting the Dynamic Weighting + Percentage Method: increase fan vote weight in early stages to boost engagement, and shift toward judge influence in later stages to ensure professionalism. The model sets fixed stage-wise weights to retain appeal while gradually reducing controversy.

Recommendations:

1. Implement the dynamic weighting + percentage model to enhance early-stage engagement and late-stage fairness.
2. If retaining original rules, apply the Bottom 2 rule only in mid-to-late stages to avoid early-stage viewership loss.

Feel free to reach out for further discussion or clarification.

Best Regards,
Team #2607943

Appendix: AI Usage Documentation

OpenAI ChatGPT 5.2

Query 1: Does the average weekly consistency calculated by our method indicate high consistency between weeks?

Output:

The "Average Weekly Consistency" metric alone does not directly indicate that the model's performance is highly consistent or stable across different weeks.

It represents the average level of the proportion of correctly classified elimination outcomes within a "typical week." This implies:

It reflects central tendency, not dispersion: A high average may mask large fluctuations between weeks. For example, the model might perform exceptionally well in four out of five weeks (consistency > 90%) but perform poorly in one particular week (consistency < 50%) due to special circumstances (e.g., data anomalies, sudden changes in participant form). Although the average might still appear acceptable, such significant week-to-week variation indicates potential issues with the model's stability.

The correct way to assess week-to-week stability is to analyze its variability: To determine whether the model's confidence or performance remains stable from one week to the next, we must look beyond the average and examine the distribution of weekly consistency values.

Key Variability Metrics and Their Interpretation:

Standard Deviation: Directly measures the average distance of each week's consistency value from the mean.

High value: Indicates that weekly results are widely spread around the mean, suggesting inconsistent performance and poor stability across weeks.

Low value: Indicates that weekly results are tightly clustered around the mean, suggesting relatively stable performance across weeks.

Coefficient of Variation (CV): This is a standardized metric calculated as (Standard Deviation / Mean). It is particularly useful for comparing the relative volatility of datasets with different means.

High CV (e.g., > 0.2 or 20%): Indicates high relative volatility. Even with a high mean, weekly consistency is unstable.

Low CV (e.g., < 0.1 or 10%): Indicates low relative volatility, suggesting highly predictable and stable week-to-week performance.

DeepSeek

Query 1: I want to know how to incorporate the preferences of both judges and fans without losing professionalism while also ensuring entertainment value. Could you please elaborate on how some domestic and international variety shows achieve this?

Output:

Good, let's delve into the fabric of these shows and see how top domestic and international variety programs use specific mechanisms to "play tricks" and achieve a balance between professionalism and entertainment. I will use a "Spectrum of Balancing Strategies" to break this down for you, arranging it from "Heavily skewed towards professionalism" to "Heavily skewed towards the masses," with a focus on analyzing the ingenious designs in the intermediate zone.

Left End of the Spectrum: Professional Authority Leads, Public Participation Assists**Case 1: The Dutch original The Voice & the Global Format (The Voice series)**

Core Mechanism: "Blind Auditions with Turning Chairs" + "Coaches/Mentor Team System."

How It Balances:

Blind Audition Stage (Pure Professionalism): Judges (coaches) make decisions based solely on voice, filtering out all non-professional factors like appearance or stage presence. This is the most bare and pure manifestation of professionalism, also creating intense drama ("I turned my chair for you!").

Battle/Knockout Rounds (Professionalism Leads): Within their respective teams, coaches have complete control over contestants' advancement. Here, coaches play dual roles as "trainer" and "judge," with their professional judgment and team strategy being the sole criteria.

Live Final Stage (Introducing the Public): Public voting (usually via app or phone) is only opened during the final live shows. By this point, those remaining are the "elite" filtered through multiple professional rounds. The public chooses from this high-quality pool, ensuring a baseline quality for the outcome.

Design Essence: "Professional filtration first, public choice later." Power resides with the coaches for most of the program. The public participates only in the final "coronation" stage, ensuring the show's musical quality while simultaneously igniting a sense of mass participation at the crucial moment.

Case 2: UK/US Strictly Come Dancing / Dancing with the Stars

Core Mechanism: "Professional Dancer Paired with Celebrity Amateur" + "Combination of Judges' Scores and Audience Votes."

How It Balances:

Clear Weight Formula: Typically, "judges' scores determine the ranking, audience votes determine who faces elimination." For example, judges give the lowest score to a pair (professionally judging them as the weakest), but the audience can save them from elimination through voting.

Professionals Set the Standard, the Public Saves: The judges' scores establish clear technical standards (footwork, timing, partnership). The audience's emotional votes (liking a

certain celebrity, being moved by a story) then act as a “corrective” or “protective” force. This directly creates the classic conflict between “professional coldness” and “public popularity,” generating substantial buzz.

Special Power: In case of a tie, the head judge usually has the final say, once again reinforcing the bottom-line role of professional authority.

Middle of the Spectrum: The Zone of Ingenious Dynamic Balance

Case 3: China’s Singer (I Am a Singer)

Core Mechanism: “500-member Mass Audience Jury” + “Expert Advisory Panel (no voting rights).”

How It Balances:

Stratified “Professionalization” of the Audience: The 500 jury members are not selected through open online casting but are recruited with strict age quotas (e.g., brackets every 10 years). This simulates an “idealized” music market, avoiding the risk of manipulation by a single fanbase, making “public preference” closer to “national preference.”

“De-powering” and “Discoursing” Professionalism: The expert advisory panel (e.g., Liang Qiaobai, Shan He) does not vote, only comments. Their role is to translate professional logic (ingenuity in arrangement, vocal difficulty) for the audience, elevating the aesthetic level of the entire program and its viewers, guiding the mass jury to appreciate and vote more “knowledgeably.”

Result Uncertainty: The audience remains unaware of the specific ranking until the final moment of announcement, maintaining extreme suspense. Professional commentary and public voting are clearly separated: professionalism influences perception but does not directly intervene in the result.

Design Essence: “Transform the public sample, strip professionals of direct power, but amplify their professional voice.”

Case 4: China’s Dance Smash

Core Mechanism: “Storm Moment” technical freeze-frame + Four-judge (panel) pass system + Introduction of public judges in later stages.

How It Balances:

Visualizing Peak Technique: The “Storm Moment” is the program’s most significant invention. It freezes and replays in slow motion, from 360 degrees, the most spectacular second of a dancer’s technique. This allows laypeople to intuitively grasp that “this move defies physics, it’s amazing!” turning the most professional aspect into a visually spectacular and shareable spectacle.

High Consistency Among Judges: Judges like Shen Wei and Shen Peiyi are towering figures in the industry. Their unanimous approval (four passes) carries immense weight. Their critiques also closely tie to the “Storm Moment” and emotional expression, giving professional comments a concrete anchor.

Phased Introduction of the Public: In later advancement stages, a “public vote channel” is introduced, but its votes usually serve as a “bonus” or a “last-minute reversal force,” not overturning the primary judgment of the professional judges.

Design Essence: “Use visuals to translate technique, letting the public first ‘see’ the professionalism, then understand it, and finally participate in selection in a limited way.”

Conclusion: A Deconstructible Toolbox

From these cases, we can extract the specific tools commonly used by program designers:

Weight Distribution Tools:

Formula: $(\text{Professional Score} \times \text{Weight}) + (\text{Public Score} \times \text{Weight}) = \text{Total Score}$.

Veto/Save Power: Judges have a “fast pass/save” privilege (e.g., the Golden Buzzer on Got Talent), or public votes can save someone eliminated by judges.

Phased Authorization: Professionals set the tone in early stages, the public decides the finale.

Participant Modification Tools:

Professionalizing the Public: e.g., Singer’s age-stratified audience jury.

Differentiating Judge Roles: Technical judge, charismatic judge, mentor-type judge, each with distinct functions.

Standard Translation Tools:

Visualizing Technique: The freeze-frame in Dance Smash, vocal pitch subtitles in Super-Vocal.

Narrativizing Terminology: Judges don’t say “Your second harmonic was good,” but rather “Your voice sounds kissed by an angel, with a holy sheen.”

Narrative Creation Tools:

Deliberately Creating Divergence: Editing disagreements between judges and public opinion into the main conflict of the show.

Showing the Decision Process: Broadcasting judges’ arguments and struggles during scoring, letting the audience understand the complexity of professional judgment.

Ultimately, the most successful programs all understand that professionalism and entertainment are not opposites. Through ingenious mechanisms, the depth of professionalism can become the substance of entertainment, and public adoration can crown professional value. They don’t aim for a static “50–50 split” but design a dynamic, fluctuating interactive process. This process itself is the most captivating drama.